THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 10

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** $Fe_2O_3 \cdot x\,H_2O$	Hydrated iron(III) oxide
2	**(b)** NH_4OH and HCl	Base + acid
3	**(b)** Student A is correct	Ethanoic acid
4	**(d)** gallium	Low melting point
5	**(c)** oxygen	Used in respiration
6	**(d)** CO_2	Brisk effervescence
7	**(a)** C_3H_8	Alkane
8	**(c)** urea	Removed by kidneys
9	**(c)** Incorrect — endocrine glands secrete directly into blood	Ductless
10	**(d)** stem tuber (eyes)	Buds on tuber
11	**(a)** anther	Stamen
12	**(a)** 9 : 3 : 3 : 1	Independent assortment
13	**(b)** TT	Homozygous
14	**(c)** real, inverted and magnified	Negative m
15	**(a)** Incorrect — violet is shorter; our eyes are more sensitive to blue	Scattering
16	**(a)** 484 Ω	$R = V^2/P$
17	**(b)** kilowatt hour	1 unit
18	**(d)** long current-carrying solenoid	Inside
19	**(a)** artificial ecosystem	Man-made
20	**(b)** biological magnification	Non-biodegradable

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	In combination, two or more substances form one; in decomposition, one substance breaks into two or more. e.g. $2\,H_2O \longrightarrow 2\,H_2 + O_2$ is the reverse of $2\,H_2 + O_2 \longrightarrow 2\,H_2O$.	1+1
22	False — dilution decreases the concentration of H_3O^+ ions per unit volume.	2
23	By forming pseudopodia that engulf food into a food vacuole, where it is digested.	2
24	Promote growth of the stem (stem elongation), seed germination.	2
25	Transfer of pollen grains from the anther to the stigma.	2
26	TT : Tt : tt = 1 : 2 : 1	2
27	Radius of the sphere of which the mirror is a part; $R = 2f$.	2
28	True — outside the magnet they go from N to S and inside from S to N.	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Gas bubbles, test tube becomes warm, zinc dissolves gradually; $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$	2+1
30	(i) sonorous (ii) good conductor, ductile (iii) non-metal that conducts electricity	1+1+1
31	Reach only cells connected by nerves; after an impulse, the cell needs time to reset; cannot continuously generate impulses	3
32	Sexually transmitted diseases: gonorrhoea, HIV-AIDS; use condoms, avoid multiple partners	1+1+1
33	Myopia; concave lens; $f = -1.2$ m ⇒ $P = -0.83$ D	1+1+1

34	Short circuit: live and neutral touch, resistance falls, huge current; overloading: too many appliances drawing more current than rated	3
35	H_3O^+ decreases; pH increases; the base neutralises the acid and finally OH^- is in excess.	1+1+1
	OR: Cooking/preserving food; raw material for NaOH, baking soda, washing soda, bleaching powder; de-icing roads	1+1+1
36	Break down dead organisms and wastes into simple substances that return to soil; bacteria, fungi	2+1
	OR: Use cloth/jute bags; reusable bottles and containers; avoid single-use plastic; recycle	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) (a) between F and C, real, inverted, diminished (b) at infinity, real, highly magnified (c) behind mirror, virtual, erect, magnified (ii) between P and F	4+1
	OR: (i) (a) at F, real, inverted, highly diminished (b) between F and 2F, real, inverted, diminished (c) at 2F, real, inverted, same size (ii) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	3+2
38	(i) incomplete combustion deposits carbon (ii) air holes are blocked (iii) butane (iv) produces poisonous CO (v) $2\,C_4H_{10} + 13\,O_2 \longrightarrow 8\,CO_2 + 10\,H_2O$	1+1+1+1+1
	OR: (i) esterification (ii) $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (iii) catalyst/dehydrating agent (iv) saponification — gives ethanol and sodium ethanoate (v) perfumes/flavouring	1+1+1+1+1
39	(i) $I = I_1 + I_2 + I_3$, $\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \Rightarrow \frac{1}{R_p} = \sum \frac{1}{R_i}$ (ii) each device gets the full voltage; one failing does not stop others; lower total resistance	3+2
	OR: (i) cell, key, ammeter, resistor (nichrome wire), rheostat in series and voltmeter across resistor; record V and I for different settings; V/I is constant (ii) straight line through origin (iii) resistance	3+1+1
40	(i) four chambers; right side receives deoxygenated blood from vena cava and pumps to lungs; left side receives oxygenated blood from pulmonary veins and pumps through aorta; valves prevent backflow (ii) blood passes once through the heart; oxygenated at gills	4+1
	OR: (i) two kidneys, ureters, urinary bladder, urethra (ii) filtration in glomerulus into Bowman's capsule; selective reabsorption of glucose, amino acids, salts, water in tubules (iii) depends on excess water and dissolved wastes; hormones regulate reabsorption	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER